

EE 115 Homework 9

Kaushik Vada

The small-signal closed-loop transfer is

$$H(s) = \frac{\Theta_o(s)}{\Theta_i(s)} = \frac{KG(s)}{s + KG(s)}, \quad \Theta_e(s) = \Theta_i(s) - \Theta_o(s) = \Theta_i(s) \frac{s}{s + KG(s)}.$$

The VCO relation (1) gives $X(s) = \frac{s \Theta_o(s)}{K_v}$.

a) $G(s) = 1$, $\theta_i(t) = \Delta\theta u(t)$.

$$\Theta_o(s) = \frac{K}{s + K} \frac{\Delta\theta}{s} = \frac{\Delta\theta K}{s(s + K)} \implies \theta_o(t) = \Delta\theta (1 - e^{-Kt}).$$

The response is a first-order rise from 0 to the final value

$$\theta_o(\infty) = \Delta\theta.$$

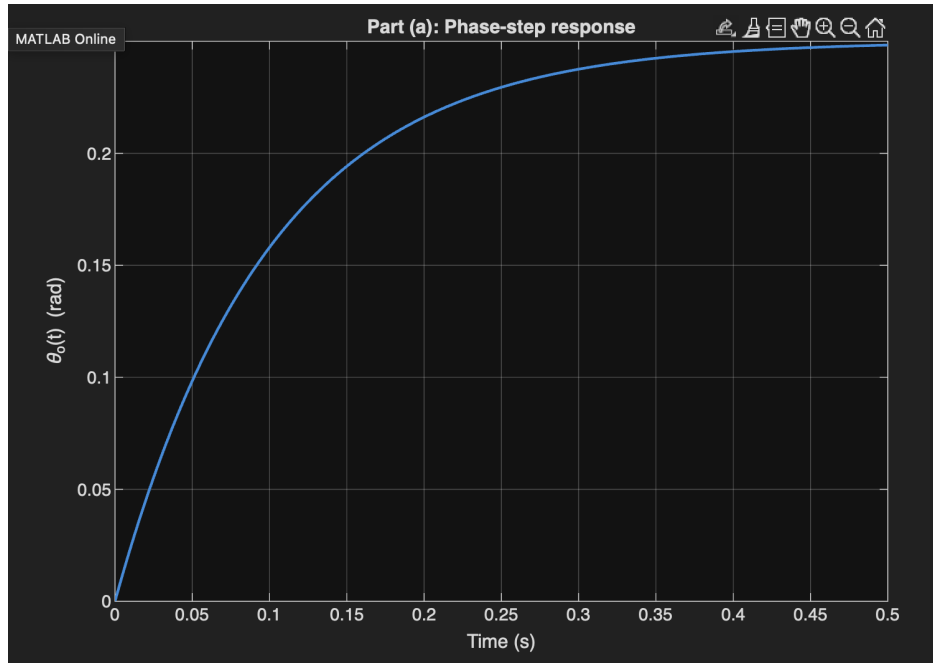


Figure 1: Part (a) simulated phase-step response.

b) $G(s) = 1$, $\dot{\theta}_i(t) = 2\pi k_f u(t) \Rightarrow \theta_i(t) = 2\pi k_f t u(t)$.

$$\Theta_i(s) = \frac{2\pi k_f}{s^2}, \quad \Theta_o(s) = \frac{K}{s + K} \frac{2\pi k_f}{s^2} = \frac{2\pi k_f K}{s^2(s + K)}.$$

Using $d\theta_o/dt = \mathcal{L}^{-1}\{s\Theta_o(s)\}$,

$$\frac{d\theta_o}{dt} = 2\pi k_f (1 - e^{-Kt}), \quad x(t) = \frac{2\pi k_f}{K_v} (1 - e^{-Kt}).$$

The phase error is

$$\theta_e(t) = \mathcal{L}^{-1}\left\{\frac{2\pi k_f}{s(s + K)}\right\} = \frac{2\pi k_f}{K} (1 - e^{-Kt}), \quad \theta_e(\infty) = \frac{2\pi k_f}{K} \propto \frac{1}{K}.$$

Larger loop gain K reduces the steady-state frequency-tracking error.

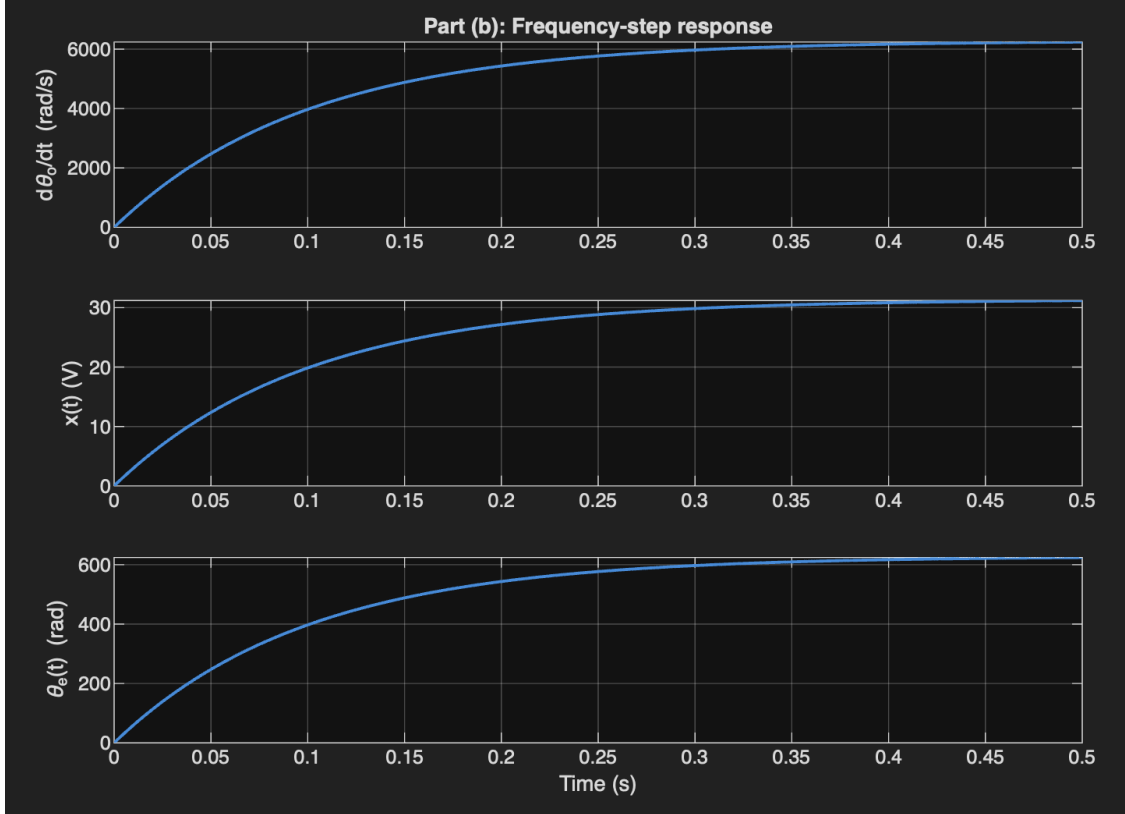


Figure 2: Part (b) simulated frequency-step response ($d\theta_o/dt$, $x(t)$, and $\theta_e(t)$).

c) $G(s) = \frac{s+a}{s}$, $\dot{\theta}_i(t) = 2\pi k_f u(t) \Rightarrow \Theta_i(s) = \frac{2\pi k_f}{s^2}$.

$$H(s) = \frac{K(s+a)}{s^2 + K(s+a)}, \quad \Theta_e(s) = \frac{2\pi k_f}{s^2 + K(s+a)}.$$

Steady-state values (final-value theorem):

$$\theta_e(\infty) = \lim_{s \rightarrow 0} s \Theta_e(s) = 0, \quad x(\infty) = \lim_{s \rightarrow 0} s X(s) = \lim_{s \rightarrow 0} \frac{2\pi k_f K(s+a)}{K_v [s^2 + K(s+a)]} = \frac{2\pi k_f}{K_v}.$$

With the additional pole-zero in $G(s)$, the loop becomes type-II and eliminates steady-state error to a frequency step.